Ruben Valenzuela

Benjamin Lacount II

Daedra minigan

Professor Viswanatha Rao

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Reflective Journal

In this reflective journal, we share our collective learning journey from Lab 04, where we built, trained, and evaluated a Convolutional Neural Network (CNN) using the MNIST dataset. This lab provided hands-on exposure to the foundational building blocks of deep learning and gave us both technical insights and personal growth. Beyond just coding, it challenged us to think critically about why certain design decisions work, how models learn from data, and how to interpret results in a meaningful way.

Learning Insights

During this lab, we discovered how CNNs specialize in processing image data by leveraging convolutional layers that act like feature detectors. Each convolutional filter scans across the input image, extracting meaningful patterns such as edges, curves, or textures. We learned that the kernel size determines the receptive field of each filter, while the number of filters controls how many unique features the network can learn simultaneously. This insight built directly on our prior knowledge of fully connected neural networks, highlighting why CNNs are far better suited for image classification tasks.

We also connected these new concepts with our existing knowledge of neural networks. Previously, we understood the importance of activation functions and loss functions, but CNNs introduced us to the idea of spatial feature preservation. Pooling layers, for example, reduce dimensionality while retaining the most important information. This gave us a clear picture of how CNNs "see" images in progressively abstract ways.

What surprised us most was the simplicity and efficiency of the MNIST dataset. Even though the images are small grayscale digits, CNNs were able to achieve extremely high accuracy. This demonstrated how deep learning models can extract incredible levels of detail from data that humans might consider basic. It also gave us confidence that with larger and more complex datasets, these techniques could scale to much harder problems such as handwriting recognition in real documents or medical image analysis.

Challenges and Growth

One challenge we faced was selecting appropriate hyperparameters, such as batch size, number of epochs, and the architecture depth. At first, we trained with too many epochs and saw the validation accuracy plateau while training accuracy kept climbing, clear evidence of overfitting. To solve this, we applied Dropout layers, which randomly deactivate neurons during training to prevent the model from memorizing the data. This not only improved generalization but also taught us the importance of regularization in real-world scenarios.

Another difficulty was fully understanding the role of one-hot encoding for labels. At first, it was not obvious why categorical cross-entropy required one-hot vectors instead of raw integers. By experimenting with both approaches, we saw that one-hot encoding allows the model to compare predicted probability distributions against the true class effectively. Reading TensorFlow/Keras documentation and testing small code snippets helped clarify this.

We also had to learn how to monitor results systematically. Visualizing loss and accuracy curves across epochs allowed us to diagnose underfitting, overfitting, and convergence behavior. These plots became crucial tools to guide our decisions. This iterative process of "experiment → observe → adjust" taught us resilience and sharpened our problem-solving skills.

Personal Development

This lab significantly changed our perception of deep learning. Before, we saw CNNs as just another type of neural network, but now we understand them as highly specialized architectures designed for vision tasks. We realized that preprocessing, architecture design, and regularization are just as important as the training process itself. CNNs are not magic, they are structured in a way that mimics how humans recognize patterns, and their success depends heavily on how we prepare the data and tune the network.

We also grew in teamwork. Dividing tasks like coding, researching documentation, and interpreting results allowed us to support one another. Whenever one of us struggled with a concept, another provided clarification, which mirrored how real-world machine learning teams collaborate.

Looking ahead, we are eager to explore deeper CNN architectures such as VGG and ResNet. These models add more convolutional layers and innovative shortcuts, which can capture even more complex features. We are also curious about how techniques like **data augmentation** (rotations, flips, noise injection) could improve generalization further, making models robust against variations in real-world inputs.

Even though we were somewhat familiar with neural networks, this lab gave us a fresh perspective: CNNs extend the foundations of dense layers to handle structured data like images with elegance and efficiency.

Key Equation: Categorical Cross-Entropy

The categorical cross-entropy loss function we used in this lab can be written as:

L = - Σ y_true * log(y_pred)

where y_true represents the true label (one-hot encoded) and y_pred represents the predicted probability distribution over the classes.

Training Visualizations

The following plots illustrate the model's accuracy and loss trends during training: